SURV701 / SURVMETH701

Analysis of Complex Sample Survey Data

Joint Program in Survey Methodology, University of Maryland

Program in Survey and Data Science, University of Michigan

Spring 2026

Abstract: Analysis of Complex Sample Survey Data covers the following topics: the development and handling of selection and other compensatory weights for survey data analysis; the effects of stratification and clustering on survey estimation and inference; alternative variance estimation procedures for estimated survey statistics; methods and computer software that take into account the effects of complex sample designs on survey estimation and inference; and methods for handling missing data, including weighting adjustment and imputation.

Instructor: **Taylor Lewis, PhD (thlewis@rti.org)**
Dates: **January 8 – April 16**
Lectures: **Thursday, 1:00 p.m. - 3:45 p.m.**
Locations: **University of Michigan, ISR 1070**
University of Maryland, LEF 2208
Zoom: **https://umich.zoom.us/j/95985553217 (passcode: 7012026)**
Office Hours: **By appointment; recommend email for quickest response**
Grader: **Maze Nead (mazenead@umich.edu)**

Course Description

Standard courses on statistical analysis assume that survey data are from a simple random sample of the target population. Little attention is given to characteristics often associated with survey data, including missing data, unequal probabilities of observation, and stratified multistage sample designs. Most standard statistical programs in software packages commonly used for data analysis (e.g., SAS®, R, and Stata®) do not allow the analyst to take most of these properties of survey data into account. Failure to do so can have an important impact on the estimation and inference for all types of analyses, ranging from simple descriptive statistics to the estimation of parameters of multivariate models.

This course provides an introduction to statistical procedures that have been developed for the analysis of complex sample survey data. Relevant design features of complex sample survey data include weights that take into account differences in probability of selection into the sample and differences in subgroup response rates, in addition to the stratification and cluster sampling employed in the multistage sampling procedures used to select households and individuals.

The course will then move on to the introduction of variance estimation techniques that have been developed to take into account the stratification and cluster sampling that are properties of the multistage sampling designs used by most major survey programs. These will initially be discussed in terms of the estimation of sampling variances for descriptive statistics: sample means, proportions and quantiles of distributions. The course will then turn to software and procedures for commonly used analyses, including testing for between-group differences in means

and proportions, linear regression analysis for continuous dependent variables, contingency table analysis for categorical data and logistic regression for categorical responses, and generalized linear models for ordinal and count data.

Prerequisites

The prerequisites for SURV/SURVMETH701 include one or more graduate courses in statistics, a course in applied sampling methods, or permission of the instructor. The course is presented at a moderately advanced statistical level. Although the course will review the fundamentals of statistical analysis methods for survey data and provide examples on the use of statistical software, it will be assumed that the students are familiar with statistical methods, including multiple regression and logistic regression. The initial lectures in the course will review the various complex features of sample designs and how they influence estimation and inference based on survey data. The course syllabus and level of instruction also assume that students are familiar with basic sampling procedures, including simple random sampling, stratification, cluster sampling and multi-stage sample designs. Students who do not have

1. experience using Excel for calculations, including logarithmic calculations (or at least how to do these calculations with a scientific calculator quickly)
2. familiarity with interpretation of mathematical symbols for common calculations included in the course
3. basic experience using SAS , Stata, or R, including opening/saving files, creating libraries, and some basic descriptive statistics commands
4. understanding of the assumptions and interpretations of linear regression and logistic regression.
5. graduate-level training in sampling techniques or graduate course in statistics

should expect to devote additional time during the first weeks of the course to supplemental readings on this topic.

Course Format

Students in the class will meet in classrooms in Ann Arbor (ISR 2070) and College Park (LEF 2208). These classrooms will be linked via a Zoom feed that will allow the students at all locations to see as well as hear the instructor and students in the other locations.

Class time will be used for a combination of lectures and discussion of examples and homework. Lecture notes and examples (those inherited from Professor Yan Li and preceding professors) will be presented on PDF slides and copies of these materials will also be available to each student on the course web site (UMich Canvas). Questions are welcomed during lectures (see more below), and discussion of the topics is encouraged.

Textbook and Class Reading

The textbook for this course will be the 3rd Edition of *Applied Survey Data Analysis* (ASDA, 2025; publisher: Chapman Hall / CRC Press) by West, Heeringa, and Berglund. Students can purchase the course text from online retailers (e.g., Amazon.com, or crcpress.com). Assigned readings will generally consist of selected sections from the chapters in the course text.

Grading

The course grade will be based on the following criteria:

Attendance (5%)
Homework Assignments (45%)
Late assignments will be penalized as follows:
- 1-2 days late: 10% off (submitted before Friday at 11:59 p.m.)
- 3-7 days late: 25% off (submitted before Wednesday at 11:59 p.m.)
- 8+ days late: no credit

In-Class (Virtual) Midterm Exam (25%)
Take-Home Final Exam (25%)

Attendance

While Professor Lewis does not formally take attendance, students are expected to attend each lecture, participate in class discussions, and complete all homework on time. Unless prior arrangements have been made, in-person learning is the strict policy of the department.

Homework Assignments

Assigned roughly biweekly and generally due the following week. Your solutions should be submitted on Canvas via the Homework Assignments tool as a single Word or PDF file. Assignments will be graded electronically and returned the week after their due dates, and the solutions will be discussed in class on that date. The course homework assignments will most often be computational exercises found at the end of chapters in the ASDA textbook. These exercises generally involve analyzing a specified survey data set and then interpreting the results of those analyses. Students are free to use whatever software procedures they would like (e.g., SAS, R, or STATA), as long as the procedures implement the methods discussed in class. The data needed for these assignments will be provided to you via the course web site (Canvas).

Students are encouraged to work in groups on the homework assignments. **However, the work that is submitted must be done by each student**. Academic integrity as outlined in the respective graduate student policies at the two Univ. (e.g. https://rackham.umich.edu/academic-policies/) will be required, and apparent violations of these policies will be dealt with on an individual basis. Assignments are due at 11:59 p.m. on Wednesdays, the night before the lecture dates, with the penalties for late submissions described above under the Grading section of the syllabus.

Tentative Schedule		
Class Date	**Topic/Homework**	**Readings**
January 8	Module 1 Survey estimation and inference for complex sample designs. Multi-stage designs, stratification, cluster sampling, finite population corrections, design effects, effective sample size. Introduction to course data sets. **Homework 1 Distributed**	Syllabus Chapters 1 and 2, ASDA
January 15	Module 2 Weighting, models and assumptions for inference from complex sample data, sampling distributions, and confidence intervals.	Chapter 3 (3.1-3.5), ASDA
January 22	Module 3 Sampling error calculation models; Ultimate clusters. **Homework 1 Due 11:59 p.m. on 1/21**	Section 4.3, ASDA
January 29	Module 4 Sampling error estimation for descriptive statistics. Taylor Series linearization method. Preparing for survey data analysis. Sampling error estimation for descriptive statistics using software. **Homework 2 Distributed**	Chapter 3 (3.6-3.8), Chapter 5 (5.1, 5.2), ASDA
February 5	Module 5 Replication Methods for Variance Estimation. Jackknife Repeated Replication (JRR). Balanced Repeated Replication (BRR). **Homework 2 Due 11:59 p.m. on 2/4** **Homework 3 Distributed**	Chapter 3 (3.6-3.8), ASDA

February 12	Module 6 Inference for percentiles and indices. Subpopulation estimates. Functions of survey estimates including differences and indices. **Homework 3 Due 11:59 p.m. on 2/11**	Chapter 5 (5.3-5.6), ASDA
February 19	Module 7 Analysis Methods for Categorical Data. Rao-Scott and related hypothesis tests. Odds ratios and relative risks.	Chapter 6, ASDA
February 26	Midterm Exam (Virtual during Class Time)	
March 5	UMich Spring Break – no class	
March 12	Review of Midterm Exam Module 8 Review of Linear Regression	Chapter 7 (7.1-7.5), ASDA
March 19	UMD Spring Break – no class	
March 26	Module 9 Linear Regression for Complex Samples **Homework 4 Distributed**	Chapter 7 (7.1-7.5), ASDA
April 2	Module 10 Logistic Regression for Complex Samples **Homework 4 Due 11:59 p.m. on 4/1** **Homework 5 Distributed**	Chapter 8 (8.1-8.8), ASDA
April 9	Module 11 Imputation Methods – Part I	Chapter 12 (12.1-12.4), ASDA

	Homework 5 Due 11:59 p.m. on 4/8	
April 16	Module 12 Imputation Methods – Part II **Final Exam Distributed** **Final Exam Due 11:59 p.m. on 4/23**	